Long-term low-calorie low-protein vegan diet and endurance exercise are associated with low cardiometabolic risk.

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BACKGROUND: Western diets, which typically contain large amounts of energy-dense processed foods, together with a sedentary lifestyle are associated with increased cardiometabolic risk. We evaluated the long-term effects of consuming a low-calorie low-protein vegan diet or performing regular endurance exercise on cardiometabolic risk factors. METHODS: In this cross-sectional study, cardiometabolic risk factors were evaluated in 21 sedentary subjects, who had been on a low-calorie low-protein raw vegan diet for 4.4 +/- 2.8 years, (mean age, 53.1 +/- 11 yrs), 21 body mass index (BMI)-matched endurance runners consuming Western diets, and 21 age- and gender-matched sedentary subjects, consuming Western diets. RESULTS: BMI was lower in the low-calorie low-protein vegan diet (21.3 +/- 3.1 kg/m(2)) and endurance runner (21.1 +/- 1.6 kg/m(2)) groups than in the sedentary Western diet group (26.5 +/- 2.7 kg/m(2)) ($p < 0.005$). Plasma concentrations of lipids, lipoproteins, glucose, insulin, C-reactive protein, blood pressure (BP), and carotid artery intima-media thickness were lower in the low-calorie low-protein vegan diet and runner groups than in the Western diet group (all $p < 0.05$). Both systolic and diastolic BP were lower in the low-calorie low-protein vegan diet group (104 +/- 15 and 62 +/- 11 mm Hg) than in BMI-matched endurance runners (122 +/- 13 and 72 +/- 9 mmHg) and Western diet group (132 +/- 14 and 79 +/- 8 mm Hg) ($p < 0.001$); BP values were directly associated with sodium intake and inversely associated with potassium and fiber intake. CONCLUSIONS: Long-term consumption of a low-calorie low-protein vegan diet or regular endurance exercise training is associated with low cardiometabolic risk. Moreover, our data suggest that specific components of a low-calorie low-protein vegan diet provide additional beneficial effects on blood pressure.